

Antoine Lorient

AI RESEARCHER, APPLIED MACHINE LEARNING FOR HUMAN AND COMPUTER INTERACTION

Paris, France

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Industry 2023 - present

Wolters Kluwer Legal Software

Paris, France

MACHINE LEARNING ENGINEER

2024 - present

- Reviewed and contributed to the architecture of an asynchronous RAG application.
- Conducted product presentations and demos from an AI perspective to several clients.
- Refactored the training and inference pipeline of our production system to facilitate onboarding.
- Defined Epic and user stories for the AI team, participated in prioritisation meetings.

Lili.ai

Paris, France

LEAD DATA SCIENTIST

2023 (6 months)

- **Designed, developed and deployed a machine learning framework in Kubernetes**, including experiment tracking (Polyaxon), data quality monitoring with human labelling (Label Studio), dataset bookkeeping (S3 and DVC) and model serving (BentoML).
- **Supervised and mentored a Big Data Master student** working with vector databases (Milvus) and LLMs (BERT-based) for similarity search and document classification.
- Advised and reviewed the overall data processing pipeline (Prefect).
- Contributed end-user features impacting both frontend and backend (Vue.js, Django).
- **Acted as a Scrum master for a team of 8 people**, organising stand-up, planning and retrospective meetings.

Research 2014 - 2022 (8 years)

CNRS - IRCAM - INRIA

Paris, France

POST-DOCTORAL RESEARCHER

Jan. 2020 - Sept. 2022 (2.5 years)

- Studied the perception of motor learning in tasks involving complex and creative movements by implementing a **metric learning** approach on top of Dynamical Time Warping (Pytorch) - published in *PLOS One*.
- Improved upon and released an **open-source project** on Github and Pypi providing the soft-DTW algorithm for CUDA devices (Pytorch, CUDA) now downloaded more than 6k times - available at *pysdtw*.
- Optimised a training schedule for a motor task using **reinforcement learning** and MABs based on a framework specialised in interactive machine learning (*Marcelle*, Javascript, Python) - published in *IHM'23*.
- Developed a prototype for movement exploration and modelling based on an algorithm performing live incremental clustering of time series (Javascript, MaxMSP, Python, C++) - demoed at *NIME'22*.

University of Glasgow

Glasgow, United Kingdom

RESEARCH ASSISTANT AND PHD CANDIDATE

Mar. 2015 - Sep. 2019 (4.5 years)

- Researched *computational methods for the study of novel gestural interactions*:
 - **Deployed discriminative models** (Keras) to create virtual surfaces from RGB-D sensors - published at *ISS'17*.
 - **Optimised gameplay experiences** for custom game controllers using user probabilistic modelling - available on *arXiv*.
 - Explored and characterised user physical capabilities with audio-based reinforcement learning.
 - **Recorded a synthetic dataset** for detecting repetitive movements from accelerometer data and **wrote from scratch a deep model** based on 1D convolutions (Python, Keras).
- Contributed to the European project MoreGrasp:
 - **Led redaction** of deliverable documents involving coordination with several project members.
 - Implemented visual feedback mechanisms on smartwatches (Android).
 - **Prepared live demos** and posters for review meetings.
 - **Established contacts** with health practitioners at Glasgow's hospital leading to workshops and a funding proposal.
- Analysed visualisation techniques (*Jupyter notebooks*) for low-dimensional embeddings, i.e. t-SNE.
- **Tutored the courses Artificial Intelligence and Data Fundamentals** for Undergraduate and Master students.
- Developed and released open source software for Intel Realsense camera. This package was recommended by Intel as a primary choice for community maintained Python bindings (until librealsense 2.0 was released) and has accrued over one hundred stars on github and 54k downloads to this day - available at *pyrealsense*.

Stockholm University, Mobile Life

Stockholm, Sweden

RESEARCH ASSISTANT

Feb. 2014 - Oct. 2014 (9 months)

- Explored innovative designs for always-on in the background speech interactions - published at *CHI'15*. **Designed and ran an experiment** capturing day-long recordings of participants' everyday life followed by qualitative analysis and design workshop with trained Interaction designers.

Industry --- 2008 - 2013 (6 years)

Now & Net

Paris, France

FOUNDER

2013 - 2015 (2 years)

- Co-founded a startup providing a collective musical experience organised around DJ-ing.
- Developed a proof-of-concept recommender system (Neo4j) for creating attractive playlists.
- Wrote R&D plans leading to the successful application to the status of JEI.

Ericsson AB

Stockholm, Beijing, Seattle

MULTIPLE POSITIONS

2008 - 2013 (6 years)

- **SCRUM master** and Software Developer [2013]: Acted as the “guardian of the process” to facilitate daily operations of a cross functional team (8 people). Animated planning meetings, daily follow-up meetings and retrospective meetings.
- **System engineer** at AT&T laboratory [summer 2012]: Integrated features into the customer network with laboratory and in-the-field experiments.
- **Research engineer** at Ericsson Research [2012]: Prepared the 4G+ prototype for the Mobile World Congress.
- **Software Developer** 4G/LTE [2008/2011]: Developed algorithms (embedded C) from standard 3GPP specifications. Extended and supported an Erlang test framework used by 50 engineers. Provided training sessions in Stockholm and Beijing.

Education ---

University of Glasgow

Glasgow, United Kingdom

PHD IN COMPUTING SCIENCE

grad. 2019

Thesis title: A Computational Approach to Gestural Interactions of the Upper Limb on Planar Surfaces.

Royal Institute of Technology (KTH)

Stockholm, Sweden

MSc IN APPLIED PHYSICS

grad. 2009

Telecom Physique Strasbourg

Illkirch-Graffenstaden, France

MSc IN ENGINEERING AND PHYSICS

grad. 2008

Skills ---

Programming	Python, C, embedded C, C++, Erlang, Javascript, $\LaTeX$, bash, make, git, GitHub, Docker
Data Modelling	Python, Pytorch, Keras, Cuda, MFlow, Pyro, Seaborn, Jupyter
ML ops	Kubernetes, Polyaxon, Label Studio, DVC, S3, BentoML
Applied Research	Scientific Writing and Review, Quantitative Analysis, Experiment Design, Ethics
Soft skills	Lean, Agile, Scrum Master, Teaching, Tutoring, English, French (native), Swedish